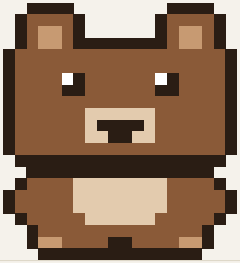
ROMA TRE · A.Y. 2026/27

Policy Evaluation Lab

Lecture 4: Randomised experiments (I)



Dr. Matteo Caruso · mcaruso4@os.uniroma3.it
Roma Tre University – Department of Business Economics
October, 2026

Where we are

- Lecture 2: subsidised firms grew 6.39 points faster, but the true effect of the grant is 3: **selection bias**
- Lecture 3: controlling for size, exports and location gives 5.45. The rest is **omitted-variable bias**, driven by management quality, which we do not observe

Today:

1. why a **lottery** removes selection bias, including the part we cannot see
2. the same grant, assigned by lottery: what our 1,500 firms would have told us
3. a real experiment on firms: does **management** matter? (Bloom et al., 2013)

PART 1

Randomised experiments



Approaches to identify causal effects

- The main challenge in evaluating a public policy, and the goal of most empirical economic research, is the elimination of **selection bias**
- . . . and to say something about the **causal effect**
- The best approach is based on a **randomised experiment**, also called a randomised controlled trial (**RCT**)
- Born in medicine (drug trials), now used by governments, international organisations and, every day, by **firms**: pilots, A/B tests, randomised promotions

Randomisation

- **Experiments**: the units eligible for a programme (people, firms, stores, plants), and who agree to take part, are divided into two random groups
  - a **treatment group**, which receives the programme, and a **control group**, which does not
- Assignment to treatment is *random*: it does not depend on the units' characteristics or choices

> Randomisation ensures that the selection bias is zero and the treatment effect is **identified**

How does it work?

Start with firms that applied for the grant.

- Suppose we have two, Alfa and Beta
- We toss a coin: heads, Alfa gets the grant; tails, Beta gets it
- We have random assignment, but it is not enough: Alfa and Beta are different (e.g. in management quality)
- If Alfa gets the grant and grows faster, it may be due to its better management, and unrelated to its lucky draw
- Hence, we need a **large sample** to ensure that differences in firm characteristics cancel out

PART 2

Random assignment



Law of Large Numbers

- By increasing the sample size, we can ensure that the sample average gets as close as we like to the population average (**Law of Large Numbers**, LLN)
- By randomly assigning firms from the same population to the treatment and control groups, we create two groups in a way that is similar to a repeated coin toss
- Therefore, we will have two groups whose average characteristics are close to those of the population
- This is true only when the groups are large enough

No selection bias

- These groups would be similar in every possible way, even in terms of **unobserved** characteristics
- E.g. we expect similar shares of exporters and of southern firms, similar size and age, and similar **management quality**
- Randomisation does not eliminate firm differences (it does not transform an Alfa into a Beta)
- It ensures that the two groups contain the same mix of firms (the two barrels contain the same proportion of Alfas and Betas)
- The treated and control groups differ only in their **treatment status** and its consequences

> Random assignment eliminates **selection bias**

Random assignment eliminates selection bias

If the sample is large enough for the LLN to work, we can replace the conditional averages with conditional expectations. With a constant treatment effect, $Y_{1i} = Y_{0i} + \tau$:

$$\begin{aligned} E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] &= E[Y_{1i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 0] \\ &= E[Y_{0i} + \tau \mid D_i = 1] - E[Y_{0i} \mid D_i = 0] \\ &= \tau + E[Y_{0i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 0] \\ &= \tau \end{aligned}$$

- Randomisation of D_i ensures that $E[Y_{0i} \mid D_i = 1] = E[Y_{0i} \mid D_i = 0]$:
- treatment and control groups come from the same population, so they are the same in every way, including Y_{0i}

Back to omitted-variable bias

$$\gamma^{short} = \gamma^{long} + \beta_{MQ} \times \delta$$

- Lecture 3: management quality was higher among subsidised firms, $\delta = 0.72$, so the omitted-variable bias was $3.82 \times 0.72 = 2.74$ points
- With a lottery, MQ is the same, on average, in both groups: $\delta \approx 0$
- The bias disappears even though we **still do not observe** MQ

> Randomisation does not need us to know what the confounders are: it balances them all

Balance check

- The first step in any evaluation is **checking for balance** between the treated and control groups
- This means comparing the average characteristics of treated and controls **before** the treatment
- Random assignment is the best way to guarantee that balance is achieved

Remember Table 2 of Lecture 2: firms chose to apply and the agency chose whom to fund, and the two groups were very different. What if the grant had been assigned by lottery?

PART 3

The grant by lottery



The same grant, assigned by lottery

- The same 1,500 firms, with the same characteristics in 2015
- The same grant, with the same true effect: $\tau = 3$ points of sales growth
- The same number of grants, 419, but the agency **draws them by lottery**

> **NOT SCIENCE FICTION**
>
> Agencies do run lotteries when funds are short and applications are many: the US H-1B visa lottery for firms' foreign hires, lotteries among eligible applicants in innovation-voucher programmes (e.g. the UK and the Netherlands), and "click days" in Italy, where funds go to the fastest applicants, a near-random rule

Balance: application vs lottery

	Grant by application (Lecture 2)		Grant by lottery	
Characteristics in 2015	Difference	t	Difference	t
Age (years)	−0.37	−0.54	1.57	2.31
Employees	8.44	6.13	1.73	1.30
Sales (thousand EUR)	2,870	7.31	303	0.96
Exporter (share)	0.24	8.53	0.01	0.25
Family-owned (share)	−0.05	−2.01	−0.01	−0.20
South (share)	0.15	5.28	0.00	−0.04
Management quality (unobserved)	*0.93*		*0.00*	

Difference = mean among grant recipients − mean among the other firms; 419 vs 1,081 firms in both cases. Simulated data. Management quality is standardised and is never in the students' datasets.

With the lottery, almost every $|t|$ is below 1.96: the two groups look alike, **including in what we cannot see**

But what about age?

- Under the lottery, the difference in age has $t = 2.31 > 1.96$: significant at 5%
- Did the lottery fail? **No**: if the true difference is zero, a test at 5% rejects 1 time in 20 by pure chance
- With 6 characteristics, one “significant” difference is not surprising

What researchers do:

- report the balance table anyway, and check that imbalances are few and small
- **control** for the unbalanced variable in the regression (Lecture 3)
- better: **stratify** the lottery, drawing winners separately within groups (e.g. by region or size class), so that key variables are balanced by design

Results: application vs lottery

Effect of the grant on sales growth (points)	**By application**	**By lottery**
Difference in means	6.39 (0.51)	**2.54** (0.52)
Regression with controls	5.45 (0.52)	**2.58** (0.50)
True effect	3	3

Standard errors in parentheses. Controls: log employees, exporter, age, South, family-owned. Simulated data.

- With the lottery, the simple difference in means is close to the truth, and **controls barely change it**: there is no bias left for them to remove
- 2.54 is not exactly 3: the 95% confidence interval, $2.54 \pm 1.96 \times 0.52 = [1.52,\ 3.56]$, contains the true value. This is **sampling variability**, not bias

PART 4

Does management matter?



Management and productivity

- Productivity differs enormously across firms, even within narrowly defined industries and countries
- Firms with better management practices (monitoring, targets, incentives, lean manufacturing) are more productive and more profitable
- But is management a **cause** of productivity, or do well-run firms simply differ in other ways (better owners, better markets, more capital)?
- This is the omitted-variable problem of Lecture 3 seen from the other side: management quality is now the **treatment**

> Bloom, Eifert, Mahajan, McKenzie and Roberts (QJE, 2013): give **management consulting** to randomly chosen plants

The experiment: firms and plants

- Large, family-owned **cotton-weaving firms** near Mumbai (India), 100–1,000 employees
- About 270 employees and $7.5 million of sales a year: large firms even by US standards
- 66 eligible firms were offered free consulting; 34 were interested; **17 firms** agreed to take part
- 28 plants, of which 20 "experimental": **14 treatment** and **6 control** plants, assigned at random (treatment and control plants never in the same firm)
- **Careful:** the 17 firms **chose** to take part; randomisation happens **within** this self-selected group

The treatment

An international consulting firm worked on **38 management practices** in four areas: quality control, inventory, factory operations, human resources

- **Diagnostic phase** (1 month): all treatment **and** control plants
- **Implementation phase** (4 months): **treatment plants only**, helping them adopt as many of the 38 practices as possible
- **Measurement phase**: data collection in all plants, weekly, for about 2 years
- Treatment plants: 733 consultant hours on average; control plants: 225
- Cost to the researchers: about $75,000 per treatment plant; at market prices, about $250,000

Balance check

	Treatment mean	Control mean	p-value of difference
Employees per firm	291	236	0.454
Employees, experimental plants	144	114	0.161
Annual sales ($m per firm)	7.06	8.37	0.598
Management score (Bloom–Van Reenen)	2.50	2.75	0.203
Share of the 38 practices adopted	0.255	0.288	0.575
Age of the plant (years)	20.5	16.8	0.662
Operating efficiency (%)	70.2	72.0	0.758
Quality (% A-grade fabric)	39.0	41.8	0.629

Source: Bloom et al. (2011), NBER WP 16658, Table 1 (selected rows). 11 treatment firms, 6 control firms.

All p-values are well above 0.10: no significant differences before the treatment. With only 17 firms, though, a balance test has **little power**: sizeable differences (291 vs 236 employees) can go undetected

Did management change?

- Before the intervention, plants had adopted between 8% and 55% of the 38 practices: even the best plant had barely half
- **Treatment plants**: from 25.6% to 63.4% of the practices, $+37.8$ percentage points
- **Control plants** (diagnostic only): much smaller increase, about 12%
- **Other plants of the treated firms** (not in the experiment): $+17.5\%$. Owners copied the practices into their other plants

> **A REVEALED-PREFERENCE TEST**
>
> Owners spread the new practices to plants nobody asked them to change: they evidently thought the practices paid off

Results

Effect of the intervention, treatment vs control plants (intention-to-treat):

- **Quality defects**: -32% (about a third)
- **Yarn inventory**: -16.4%
- **Output**: $+5.4\%$ (significant only at the 11% level)
- **Productivity**: $+11\%$ in the 2011 working paper; $+17\%$ in the first year in the published version
- **Profits**: about $230,000 a year per firm, against implementation costs below $3,000: roughly a 90% one-year return even on the market price of the consulting

Source: Bloom et al. (2011), Sections IV–V; Bloom et al. (2013), QJE 128(1), abstract.

Inference with 20 plants

- 20 plants in 17 firms is a **very small sample**: the usual large-sample t-tests may be unreliable
- The authors use two strategies:
  - about 100 weeks of data per plant (many observations over time)
  - **permutation tests**: compute the effect for **every possible** random assignment of 11 treatment firms out of 17 ($\binom{17}{11} = 12{,}376$ assignments), and ask how extreme the actual one is
- If the actual effect lies in the top or bottom 2.5% of these, it is significant at 5%

> The lottery itself provides the benchmark: **what if a different set of plants had won?**

Why didn't firms adopt these practices before?

- **Lack of information**: firms were simply not aware of many practices (about 15% of practices at the start)
- **Incorrect information**: firms knew a practice but thought it would not pay off for them (over 45% of initial non-adoption). E.g. machines were repaired when they broke down, rather than maintained regularly
- **Little competition**: barriers to entry and growth meant badly managed firms were not driven out of the market

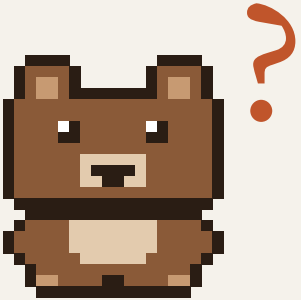
A POLICY QUESTION

If information is the barrier, should governments subsidise consulting for SMEs, as Italy did with vouchers for innovation managers? How would you evaluate such a scheme?

Lessons

- **For evaluation**: randomisation balances observed **and unobserved** characteristics; the simple difference in means becomes the causal effect
- **For managers**: the textile owners started by **piloting** the easiest changes in one part of the factory, then rolled them out once they saw results. A pilot with a control group is an experiment
- **Limits**: small samples, firms that select into the experiment, effects that may fade once the consultants leave

Next lecture: what can go wrong in an experiment (firms that do not comply, firms that drop out), intention-to-treat, and A/B testing inside companies



Questions?

mcaruso4@os.uniroma3.it · https://pel.carusomatteo.it